

# Chapter 11: Probability

Mixed revision of all 5 lessons · 13 questions

**NSW syllabus outcome**

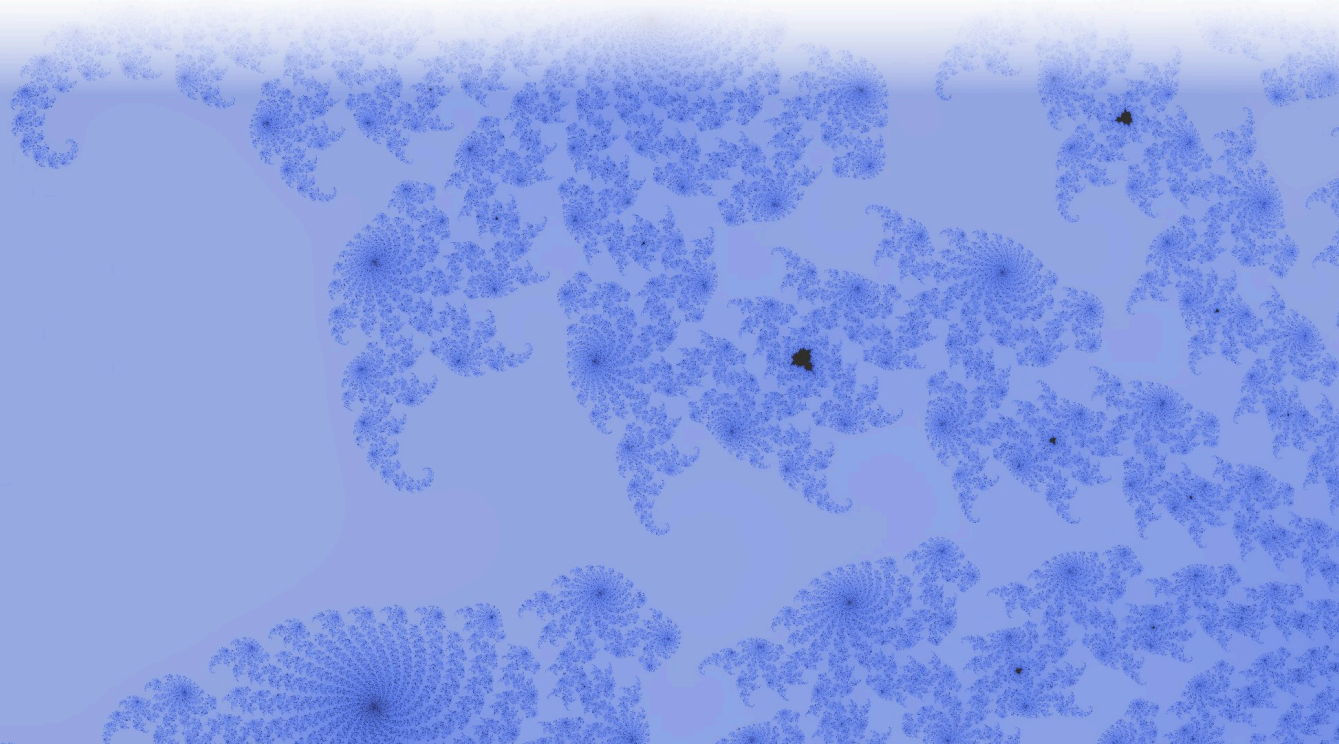
Stage 4 Statistics and probability: MA4-PRO-C-01 solves problems involving the probabilities of simple chance experiments.

NAME

CLASS

CHAPTER

**11**



**Mixed revision homework.** These questions mix all the lessons in Chapter 11, from **Easy** to **Challenging**. Each one is tagged with its lesson, so you can look back at that lesson's summary if you get stuck. Show your working in the boxes.

**YOU DO** On your own: mixed revision.

QUESTIONS 1–10 OF 13

- 1** **Easy** How many outcomes are in the sample space?  
choosing a day of the week Lesson 11.01

- 2** **Easy** Find the expected number.  
heads in 90 tosses Lesson 11.04

- 3** **Easy** Write the chance word for this probability.  
0.95 Lesson 11.03

- 4** **Easy** A bag has 3 red, 5 blue and 2 green counters. Find the probability.  
P(blue or green) Lesson 11.02

- 5** **Medium** How many outcomes are there when rolling a die? Lesson 11.01

- 6** **Medium** How many 3s are expected in 120 rolls? Lesson 11.04

- 7** **Medium** Which word describes  $P = 0.9$ ? Lesson 11.03

- 8** **Medium** Find each probability as a fraction and a percentage (1 d.p.).  
15 dark, 13 milk, 12 white chocolates: P(white) Lesson 11.02

- 9** **Challenging** Are the outcomes of a test (pass/fail) equally likely? Explain. Lesson 11.01

- 10** **Challenging** A 3 came up 25 times in 120 rolls. Find the observed P(3). Lesson 11.04

**YOU DO**

On your own: mixed revision.

QUESTIONS 11–13 OF 13

11

**Challenging**

Problems.

A bag has 12 counters and  $P(\text{green}) = \frac{1}{4}$ . How many are green? Lesson 11.02

12

**Challenging**

Problems.

Explain why  $P(E) + P(\text{not } E) = 1$ . Lesson 11.05

13

**Challenging**

Mark each event on the scale.

Describe an event that is: (a) impossible (b) unlikely (c) certain. Lesson 11.03

**YOU DO**

On your own: extra working space. Write the question number next to your working.